```sql
Use COSMOS;



DROP TABLE IF EXISTS #SC_retin --94
SELECT DISTINCT DiagnosisKey,
                TerminologyConceptKey,
                [Value],
                DisplayString, -- dd.Name ,NameAndCode,GroupedNameAndCode
                CASE
                    WHEN DisplayString LIKE '%right eye%' THEN 'RT'
                    WHEN DisplayString LIKE '%left eye%' THEN 'LT'
                    WHEN DisplayString LIKE '%bilateral%' THEN '50'
                    WHEN DisplayString LIKE '%unspecified eye%' THEN 'Unspecified'
                    ELSE '999' -- Use 999 or another code for cases with no laterality info
                END AS Laterality

INTO #SC_retin
                FROM DiagnosisTerminologyDim
                --left join DiagnosisDim dd
                -- on dd.DiagnosisKey=DiagnosisTerminologyDim.DiagnosisKey
                WHERE [Type]=N'ICD-10-CM'
                    AND (
                        --(([Value] LIKE 'E10.3%' AND [Value] != 'E10.3') AND [Value] NOT LIKE 'E10.3[6-9]%') OR
                        --(([Value] LIKE 'E11.3%' AND [Value] != 'E11.3') AND [Value] NOT LIKE 'E11.3[6-9]%') OR
                        [Value] LIKE 'H36.81%' OR
                        [Value] LIKE 'H36.82%'
                ) ;




DROP TABLE IF EXISTS #all_sc_pt_all_dx; --
with alldx as(
select scdx.PatientDurableKey,
       retin.Laterality,
       retin.Date_retin_dx,
       retin.Age_retin_dx,
       retin.icd_retin_dx,
       retin.DisplayString_retin_dx,
       retin.Insurance_retin_dx

from (select * from PROJECTS.ProjectD821712.dbo.SC_pt ) as scdx
join (
    select
        Diagnoses2.PatientDurableKey,
        --Diagnoses2.StartDateKey,
        DiagnosisDate.DateValue as Date_retin_dx,
        age.Years as Age_retin_dx,
        D.[Value] as icd_retin_dx,
        D.DisplayString as DisplayString_retin_dx,
        D.Laterality,
        ins.PrimaryCoverageFinancialClass_X as Insurance_retin_dx
        --ROW_NUMBER() OVER(
        --                  PARTITION BY Diagnoses2.PatientDurableKey, D.[Value]
        --                  ORDER BY Diagnoses2.StartDateKey ASC) AS RowNum_retin
    from DiagnosisEventFact Diagnoses2
    INNER JOIN ( SELECT * FROM #SC_retin ) D
            ON D.DiagnosisKey = Diagnoses2.DiagnosisKey
    LEFT JOIN EncounterFact ins
          ON Diagnoses2.EncounterKey = ins.EncounterKey
        LEFT JOIN DateDim DiagnosisDate ON Diagnoses2.StartDateKey = DiagnosisDate.DateKey
        LEFT JOIN DurationDim age ON Diagnoses2.AgeKey = age.DurationKey

    WHERE Diagnoses2.StartDateKey > 0
          AND Diagnoses2.[Type] <> N'Problem List'
    ) as retin
on scdx.PatientDurableKey=retin.PatientDurableKey
--where retin.Strt_date_retin>scdx.StartDateKey
)
select distinct *
into #all_sc_pt_all_dx
from alldx;



-- try to reduce unspecified dx

DROP TABLE IF EXISTS #clean_sc_retin_dx;
-- STEP 1: Normalize and add severity rank
WITH ranked_dx AS (
    SELECT
        PatientDurableKey,
        icd_retin_dx,
        LEFT(icd_retin_dx, 6) AS icd_root,
        Laterality AS original_laterality,
        Date_retin_dx,
        CASE
          --WHEN icd_retin_dx = 'H36.823' THEN 7
          --WHEN icd_retin_dx IN ('H36.821', 'H36.822') THEN 6
          --WHEN icd_retin_dx = 'H36.829' THEN 5
          --WHEN icd_retin_dx = 'H36.813' THEN 4
          --WHEN icd_retin_dx IN ('H36.811', 'H36.812') THEN 3
          --WHEN icd_retin_dx = 'H36.819' THEN 2
          --ELSE 1
          WHEN icd_retin_dx IN ('H36.823','H36.821', 'H36.822') THEN 4 --PDR
          WHEN icd_retin_dx IN ('H36.813','H36.811', 'H36.812') THEN 3 --NPDR
          WHEN icd_retin_dx = 'H36.829' THEN 2
          WHEN icd_retin_dx = 'H36.819' THEN 1
          ELSE 0
        END AS severity_rank
    FROM #all_sc_pt_all_dx
),
-- STEP 2: Keep only the most severe dx per patient + date
most_severe_dx AS (
    SELECT *
    FROM (
        SELECT *,
               ROW_NUMBER() OVER (
                   PARTITION BY PatientDurableKey, Date_retin_dx
                   ORDER BY severity_rank DESC
               ) AS rn
        FROM ranked_dx
    ) t
    WHERE rn=1
)
```

```sql
),
-- STEP 4: Known values for window inference
known_laterality_window AS (
    SELECT
        PatientDurableKey,
        icd_retin_dx,
        icd_root,
        Date_retin_dx,
        original_laterality
    FROM most_severe_dx
    WHERE original_laterality IN ('LT', 'RT', '50')
),

-- STEP 5: ±180 day inference if consistent
window_inference AS (
    SELECT
        u.PatientDurableKey,
        u.icd_retin_dx,
        u.icd_root,
        u.Date_retin_dx,
        --COUNT(DISTINCT k.original_laterality) AS distinct_laterality_count,
        k.original_laterality AS inferred_laterality_window,
          ROW_NUMBER() OVER (
                   PARTITION BY u.PatientDurableKey, u.Date_retin_dx
                   ORDER BY ABS(DATEDIFF(DAY, u.Date_retin_dx, k.Date_retin_dx))  DESC
               ) AS rn
    FROM (select * from most_severe_dx where original_laterality = 'Unspecified') u
    LEFT JOIN (select * from most_severe_dx where  original_laterality IN ('LT', 'RT', '50')) k
      ON u.PatientDurableKey = k.PatientDurableKey
      --AND u.icd_retin_dx = k.icd_retin_dx
     AND u.icd_root = k.icd_root
     AND k.Date_retin_dx BETWEEN DATEADD(DAY, -90, u.Date_retin_dx)
                             AND DATEADD(DAY,  90, u.Date_retin_dx)
    --WHERE u.original_laterality = 'Unspecified'
    --GROUP BY u.PatientDurableKey, u.icd_retin_dx, u.icd_root, u.Date_retin_dx
    --HAVING COUNT(DISTINCT k.original_laterality) = 1
),
--select * from window_inference where inferred_laterality_window is not null order by PatientDurableKey,Date_retin_dx
-- STEP 6: Final result
final_output AS (
    SELECT
        m.PatientDurableKey,
        m.icd_retin_dx,
        m.icd_root,
        m.Date_retin_dx,
        m.original_laterality,
        COALESCE(
            w.inferred_laterality_window,
            m.original_laterality
        ) AS final_laterality,
        CASE
            WHEN m.original_laterality = 'Unspecified'
                 AND (w.inferred_laterality_window IS NOT NULL)
            THEN 1
            ELSE 0
        END AS was_inferred
    FROM most_severe_dx m
    LEFT JOIN (select * from window_inference where rn=1) w
      ON m.PatientDurableKey = w.PatientDurableKey
     AND m.icd_retin_dx = w.icd_retin_dx
     AND m.Date_retin_dx = w.Date_retin_dx
),
-- Build a patient + icd_root level summary of known laterality
known_laterality_summary AS (
    SELECT
        PatientDurableKey,
        icd_root,
        -- Flag if Bilateral exists
        MAX(CASE WHEN final_laterality = '50' THEN 1 ELSE 0 END) AS has_bilateral,

        -- Count of LT and RT for mode fallback
        SUM(CASE WHEN final_laterality = 'LT' THEN 1 ELSE 0 END) AS lt_count,
        SUM(CASE WHEN final_laterality = 'RT' THEN 1 ELSE 0 END) AS rt_count
    FROM final_output
    WHERE final_laterality IN ('LT', 'RT', '50')
    GROUP BY PatientDurableKey, icd_root
),
--Merge with rows where final_laterality is still 'Unspecified'
final_with_fallback AS (
    SELECT
        f.*,
        k.has_bilateral,
        k.lt_count,
        k.rt_count,

        CASE
            WHEN f.final_laterality = 'Unspecified' AND k.has_bilateral = 1 THEN '50'
            WHEN f.final_laterality = 'Unspecified' AND k.lt_count > k.rt_count THEN 'LT'
            WHEN f.final_laterality = 'Unspecified' AND k.rt_count > k.lt_count THEN 'RT'
            WHEN f.final_laterality = 'Unspecified' AND k.rt_count = k.lt_count AND k.rt_count > 0 THEN 'LT'  -- Arbitrary tie-break
            ELSE f.final_laterality
        END AS final_laterality_final,

        CASE
            WHEN f.final_laterality = 'Unspecified' AND (
                    k.has_bilateral = 1 OR k.lt_count > 0 OR k.rt_count > 0
                )
            THEN 1
            ELSE f.was_inferred
        END AS was_inferred_final

    FROM final_output f
    LEFT JOIN known_laterality_summary k
      ON f.PatientDurableKey = k.PatientDurableKey
     AND f.icd_root = k.icd_root
)
--select top 1000 s.*
--from final_with_fallback s
--join (select PatientDurableKey from final_with_fallback
--where original_laterality='Unspecified' and final_laterality_final<>'Unspecified'
----group by PatientDurableKey
----having count(*)>3
--) multi_pt
--on multi_pt.PatientDurableKey=s.PatientDurableKey
--order by s.PatientDurableKey,Date_retin_dx

-- Final cleaned dataset
```

```sql
SELECT     PatientDurableKey,
    icd_retin_dx,
    icd_root,
    Date_retin_dx,
    original_laterality,
    final_laterality_final AS final_laterality,
    was_inferred_final AS was_inferred
into #clean_sc_retin_dx
FROM final_with_fallback;




DROP TABLE IF EXISTS #clean_sc_retin_dx_expanded;
--reduce daily insurance
WITH insurance_ranked AS (
    SELECT
        PatientDurableKey,  Laterality,        Date_retin_dx,        icd_retin_dx,      Age_retin_dx,       DisplayString_retin_dx,        Insurance_retin_dx,
        -- Assign ranks: lower = better
        CASE
            WHEN Insurance_retin_dx IN ('Medicaid', 'Medicare') THEN 1
            WHEN Insurance_retin_dx = 'Miscellaneous/Other' THEN 2
            WHEN Insurance_retin_dx = 'Self-Pay' THEN 3
            WHEN Insurance_retin_dx = '*Not Applicable' THEN 4
            WHEN Insurance_retin_dx = '*Masked' THEN 5
            WHEN Insurance_retin_dx = '*Unspecified' THEN 6
            ELSE 7  -- Just in case any unexpected value
        END AS insurance_rank
    FROM #all_sc_pt_all_dx a  -- replace with your actual table
),

best_insurance_per_day AS (
    SELECT *
    FROM (
        SELECT *,
               ROW_NUMBER() OVER (
                   PARTITION BY PatientDurableKey,Laterality, Date_retin_dx, icd_retin_dx
                   ORDER BY insurance_rank ASC
               ) AS rn
        FROM insurance_ranked
    ) t
    WHERE rn = 1
),
final_dx as (
select  a.PatientDurableKey,        a.icd_retin_dx,     a.Date_retin_dx,        a.original_laterality,      a.final_laterality as Laterality,       a.was_inferred,     b.Age_retin_dx,b.DisplayS
from #clean_sc_retin_dx a
join best_insurance_per_day b
on a.PatientDurableKey=b.PatientDurableKey and a.original_laterality=b.Laterality and a.Date_retin_dx=b.Date_retin_dx and a.icd_retin_dx=b.icd_retin_dx
),
expanded_DX as(
    SELECT
        PatientDurableKey,      Date_retin_dx,      Age_retin_dx,       icd_retin_dx,       DisplayString_retin_dx,     Insurance_retin_dx,     'RT' AS Laterality,was_inferred  -- Replace '50'
    FROM
        final_dx
    WHERE
        Laterality = '50'  -- Select bilateral cases

    UNION ALL
    SELECT
         PatientDurableKey,     Date_retin_dx,      Age_retin_dx,       icd_retin_dx,       DisplayString_retin_dx,     Insurance_retin_dx,         'LT' AS Laterality,was_inferred  -- Replace '5
    FROM
        final_dx
    WHERE
        Laterality  = '50'  -- Same bilateral cases

    UNION ALL
    -- Include the original rows where Laterality is not '50'
    SELECT
         PatientDurableKey,     Date_retin_dx,      Age_retin_dx,       icd_retin_dx,       DisplayString_retin_dx,     Insurance_retin_dx,     Laterality ,was_inferred -- Keep original Lateral
    FROM
        final_dx
    WHERE
        Laterality <> '50' --AND  p.Laterality <> 'Unspecified'
)
select *
into #clean_sc_retin_dx_expanded
from expanded_DX
;

/*

select COUNT_BIG(*) from #clean_sc_retin_dx_expanded; --27691
select  COUNT_BIG(distinct PatientDurableKey) from #clean_sc_retin_dx_expanded; --5240
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #clean_sc_retin_dx_expanded; --9266 NB includes unspecified.

select Laterality, count(Laterality) as N from  #clean_sc_retin_dx_expanded group by Laterality;
Unspecified 1946
RT  12906
LT  12839


select was_inferred, count(was_inferred) as N from  #clean_sc_retin_dx_expanded group by was_inferred;
0   24732
1   2959
*/


-- get FU
DROP TABLE IF EXISTS #clean_sc_retin_dx_expanded_enc;
with first_retin_dx as (
select PatientDurableKey, min(Date_retin_dx) as min_Date_retin_dx
from #clean_sc_retin_dx_expanded
group by PatientDurableKey
),
last_opt_enc as (
select pt.PatientDurableKey,
       MAX(dt.DateValue) AS LastEncounterDate_opt,
       COUNT(dt.DateValue) as EncounterCountAfterFirstDate_opt
from first_retin_dx pt
join EncounterFact ef
 on ef.PatientDurableKey=pt.PatientDurableKey
JOIN DepartmentDim dept
     ON ef.DepartmentKey = dept.DepartmentKey
JOIN DateDim dt
     ON dt.DateKey=ef.DateKey

where dt.DateValue>pt.min_Date_retin_dx
```

```
and
dept.DepartmentSpecialty like N'%ophth%' or
dept.DepartmentSpecialty like N'%optom%'
group by pt.PatientDurableKey
),
all_enc as (
select pt.PatientDurableKey,
       MAX(dt.DateValue) AS LastEncounterDate,
       COUNT(dt.DateValue) as EncounterCountAfterFirstDate
from first_retin_dx pt
join EncounterFact ef
 on ef.PatientDurableKey=pt.PatientDurableKey
JOIN DepartmentDim dept
      ON ef.DepartmentKey = dept.DepartmentKey
JOIN DateDim dt
      ON dt.DateKey=ef.DateKey
where dt.DateValue>pt.min_Date_retin_dx
group by pt.PatientDurableKey
)
select pt.*,
     opt.EncounterCountAfterFirstDate_opt,
     opt.LastEncounterDate_opt,
     e.EncounterCountAfterFirstDate,
     e.LastEncounterDate,
     DATEDIFF(DAY, Date_retin_dx, LastEncounterDate_opt) AS FU_day_opt,
     DATEDIFF(DAY, Date_retin_dx, LastEncounterDate) AS FU_day

into #clean_sc_retin_dx_expanded_enc
from #clean_sc_retin_dx_expanded pt
left join last_opt_enc opt
    on opt.PatientDurableKey=pt.PatientDurableKey
left join all_enc e
    on e.PatientDurableKey=pt.PatientDurableKey
    ;

select count_big(distinct PatientDurableKey) from #clean_sc_retin_dx_expanded_enc where FU_day >=180 -- 4268 3975  3419

                                                                    /*

                                                                    INSERT INTO PROJECTS.ProjectD821712.dbo.SC_retin_dx
                                                                    select * from #clean_sc_retin_dx_expanded_enc;

                                                                    */
--select count(*) from  PROJECTS.ProjectD821712.dbo.SC_retin_dx ; -- 27691
-------------------------------------------------------------------------------------------------------------------

DROP TABLE IF EXISTS #cohort;
select *
into #cohort
from PROJECTS.ProjectD821712.dbo.SC_retin_dx where FU_day >=180;


--with Loss_FU as (
--select *
--from PROJECTS.ProjectD821712.dbo.SC_retin_dx where FU_day <180
--)
----select  COUNT_BIG(distinct PatientDurableKey) from Loss_FU; --2618
--select    COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from Loss_FU; --4553


--Check 633820807   why my RT missing for two bilat dx



DROP TABLE IF EXISTS #first_sc_retin_dx;
with ordered as(
select *,
      ROW_NUMBER() OVER (
          PARTITION BY PatientDurableKey, Laterality
          ORDER BY Date_retin_dx ASC
      ) AS rn
from #cohort
),
dup_lat_pt_ordered as (
select * from
(
select *,
      ROW_NUMBER() OVER (
          PARTITION BY PatientDurableKey
          ORDER BY Date_retin_dx ASC
      ) AS rn_date
from ordered
where rn=1 and PatientDurableKey in (select PatientDurableKey from ordered where rn=1 group by PatientDurableKey having count(*)>2) --pt with unspec, LT and RT
) l
where ( rn_date=1 and Laterality='Unspecified') or (rn_date=3 and Laterality='Unspecified')
),
dup_lat_pt as(
select * from ordered
where PatientDurableKey in (select PatientDurableKey from dup_lat_pt_ordered) and rn=1
),
--select * from dup_lat_pt where PatientDurableKey='3144217' --ORDER BY PatientDurableKey,Date_retin_dx
imputed_unspec as(
 SELECT
       PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       'LT' AS Laterality,
       1 as was_inferred
    FROM dup_lat_pt     where Laterality='Unspecified'

      UNION ALL

     select
       PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       'RT' AS Laterality,
       1 as was_inferred
    FROM dup_lat_pt where Laterality='Unspecified'

    UNION ALL
```

```
    select
       PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
    FROM dup_lat_pt  where Laterality<>'Unspecified'
),
--select* from imputed_unspec where PatientDurableKey='3144217'
ranked_imputed as(
select *,
ROW_NUMBER() OVER (
          PARTITION BY PatientDurableKey, Laterality
          ORDER BY Date_retin_dx ASC
      ) AS rn
from imputed_unspec
),
final as(
select
       PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
 from ranked_imputed where rn=1

 UNION ALL
 select
       PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
                    WHEN DisplayString LIKE '%left eye%' THEN 'LT'
                    WHEN DisplayString LIKE '%bilateral%' THEN '50'
                    WHEN DisplayString LIKE '%unspecified eye%' THEN 'Unspecified'
                    ELSE '999' -- Use 999 or another code for cases with no laterality info
                END AS Laterality

INTO #complication_dx
                FROM DiagnosisTerminologyDim
                --left join DiagnosisDim dd
                -- on dd.DiagnosisKey=DiagnosisTerminologyDim.DiagnosisKey
                WHERE [Type]=N'ICD-10-CM'
                    AND (
                        --(([Value] LIKE 'E10.3%' AND [Value] != 'E10.3') AND [Value] NOT LIKE 'E10.3[6-9]%') OR
                        --(([Value] LIKE 'E11.3%' AND [Value] != 'E11.3') AND [Value] NOT LIKE 'E11.3[6-9]%') OR
                        [Value] LIKE 'H43.1%' OR
                        [Value] LIKE 'H33.4%'
                ) ;

DROP TABLE IF EXISTS #DxEvents_all; --1785
    SELECT
        diag.PatientDurableKey,
        diag.EncounterKey,
        diag.DiagnosisKey,
        diag.[Type],
        diag.[Status],
        diag.Chronic,
        diag.IsPrimary,
        age.Years,
        DiagnosisDate.DateValue,
        dtd.Name,
        rt.[Value],
        rt.DisplayString,
        rt.Laterality,
        ef.PrimaryCoverageFinancialClass_X
    into #DxEvents_all
    FROM DiagnosisEventFact diag
    INNER JOIN (select * from #first_sc_retin_dx) pt on pt.PatientDurableKey=diag.PatientDurableKey
    INNER JOIN DateDim DiagnosisDate ON DiagnosisDate.DateKey = diag.StartDateKey
    INNER JOIN DiagnosisDim dtd ON diag.DiagnosisKey = dtd.DiagnosisKey
    INNER JOIN #complication_dx rt ON diag.DiagnosisKey = rt.DiagnosisKey
    INNER JOIN EncounterFact ef ON diag.EncounterKey = ef.EncounterKey
    INNER JOIN DurationDim age ON diag.AgeKey = age.DurationKey
    WHERE diag.StartDateKey > 0
      --AND age.Years >= 18
      AND  diag.[Type] <> N'Problem List';

      /*
select top 1000* from #DxEvents_all order by PatientDurableKey,DateValue

select Laterality, count(Laterality) as N from  #DxEvents_all group by Laterality;
Unspecified 1365
50  2016
RT  16841
LT  15244


select Laterality, count(Laterality) as N from  #DxEvents_all_expanded group by Laterality;
Unspecified 1365
RT  18857
LT  17260
*/


DROP TABLE IF EXISTS #DxEvents_all_expanded; --1785
WITH InitialRTDx AS (
   select * from #DxEvents_all
),
expanded_DX as(
    SELECT
        p.PatientDurableKey,
        p.DateValue,
        p.[Value],
        --p.[Type],
     --   p.[Status],
```

```
     --   p.Chronic,
     --   p.IsPrimary,
        p.Years,
        p.DisplayString,
        p.PrimaryCoverageFinancialClass_X as CoverageFinancialClass,
        'RT' AS Laterality  -- Replace '50' with 'RT' for right eye
    FROM
        InitialRTDx p
    WHERE
        p.Laterality = '50'  -- Select bilateral cases

    UNION ALL

    SELECT
        p.PatientDurableKey,
        p.DateValue,
        p.[Value],
        --p.[Type],
     --   p.[Status],
     --   p.Chronic,
     --   p.IsPrimary,
        p.Years,
        p.DisplayString,
        p.PrimaryCoverageFinancialClass_X as CoverageFinancialClass,
        'LT' AS Laterality  -- Replace '50' with 'LT' for left eye
    FROM
        InitialRTDx p
    WHERE
        p.Laterality  = '50'  -- Same bilateral cases

    UNION ALL

    -- Include the original rows where Laterality is not '50'
    SELECT
        p.PatientDurableKey,
        p.DateValue,
        p.[Value],
        --p.[Type],
     --   p.[Status],
     --   p.Chronic,
     --   p.IsPrimary,
        p.Years,
        p.DisplayString,
        p.PrimaryCoverageFinancialClass_X as CoverageFinancialClass,
        --p.Laterality  -- Keep original Laterality for non-bilateral cases
        case
            when p.Laterality in ('Right') then 'RT'
            when p.Laterality in ('Left') then 'LT'
            else p.Laterality --Unspecfied
        end as Laterality
    FROM
        InitialRTDx p
    WHERE
        p.Laterality <> '50' --AND  p.Laterality <> 'Unspecified'
)
select *
into #DxEvents_all_expanded
from expanded_DX




DROP TABLE IF EXISTS #Dx_exclusion;
with MYPatients as(
select * from #first_sc_retin_dx
),
expanded_DX as(
select * from #DxEvents_all_expanded
),
matched_laterality as (
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY DE.DateValue ASC) as rn_firstDx

from (select * from MYPatients ) pt --where Laterality<>'Unspecified'
inner join expanded_DX DE
    on DE.PatientDurableKey=pt.PatientDurableKey and DE.Laterality=pt.Laterality
    where DE.DateValue<=pt.Date_retin_dx
    ),

match_remainder as (
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY ABS(DATEDIFF(DAY, pt.Date_retin_dx, DE.DateValue)) ASC) as rn_firstDx

from ( select * from MYPatients mp
where not exists (
    select 1
    from matched_laterality ml
    where  ml.PatientDurableKey= mp.PatientDurableKey and ml.Laterality=mp.Laterality
        )

    ) pt
inner join ( select * from expanded_DX where Laterality='Unspecified') DE --where Laterality='Unspecified'
    on DE.PatientDurableKey=pt.PatientDurableKey
    where DE.DateValue<=pt.Date_retin_dx
),

Match_remainder_unspec as(
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
```

```
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY ABS(DATEDIFF(DAY, pt.Date_retin_dx, DE.DateValue)) ASC) as rn_firstDx

from ( select * from MYPatients mp
where not exists (
    select 1
    from matched_laterality ml
    where  ml.PatientDurableKey= mp.PatientDurableKey and ml.Laterality=mp.Laterality
        )
    and not exists (
    select 1
    from match_remainder m2
    where  m2.PatientDurableKey= mp.PatientDurableKey and m2.Laterality=mp.Laterality
    )
    ) pt
inner join ( select * from expanded_DX ) DE --where Laterality='Unspecified'
    on DE.PatientDurableKey=pt.PatientDurableKey
    where DE.DateValue<=pt.Date_retin_dx and pt.Laterality='Unspecified'

),
joined_data as (
select * from matched_laterality where rn_firstDx=1

union all

select * from match_remainder where rn_firstDx=1


union all

select * from Match_remainder_unspec where rn_firstDx=1

)
select *,
       DATEDIFF(DAY, Date_retin_dx, DateValue) as Dx_intervention_DaysDifference
into #Dx_exclusion
from joined_data
;


/*
select * from #Dx_exclusion  order by PatientDurableKey

select COUNT_BIG(*) from #Dx_exclusion; --902
select  COUNT_BIG(distinct PatientDurableKey) from #Dx_exclusion; --703
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #Dx_exclusion; --902

select Laterality, count(Laterality) as N from  #Dx_exclusion group by Laterality;
Unspecified 95
RT  399
LT  408
*/

DROP TABLE IF EXISTS #final_cohort; -- remove FU and dx exlcusions
select *
into #final_cohort
from #first_sc_retin_dx dx
where not exists (
    select 1
    from #Dx_exclusion e
    where  e.PatientDurableKey= dx.PatientDurableKey and e.Laterality=dx.Laterality
        );

        /*
select COUNT_BIG(*) from #final_cohort; --6463
select  COUNT_BIG(distinct PatientDurableKey) from #final_cohort; --3904
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #final_cohort; --6463

select Laterality, count(Laterality) as N from  #final_cohort group by Laterality;
Unspecified 492
RT  3002
LT  2969


select COUNT_BIG(*) from #first_sc_retin_dx; --7365
select  COUNT_BIG(distinct PatientDurableKey) from #first_sc_retin_dx; --4268
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #first_sc_retin_dx; --7365

select Laterality, count(Laterality) as N from  #first_sc_retin_dx group by Laterality;
Unspecified 587
RT  3401
LT  3377

select count(distinct p.PatientDurableKey) from #final_cohort p --339 patients had one eye removed.
join #Dx_exclusion e
 on e.PatientDurableKey=p.PatientDurableKey  --and  e.Laterality=p.Laterality


 check matches with unspecs
 select count(*) from #Dx_exclusion where Laterality<>'Unspecified' and Laterality_dx='Unspecified' --87
 select count(*)  from #Dx_exclusion where Laterality='Unspecified' and Laterality_dx<>'Unspecified' --70
 select count(*)  from #Dx_exclusion where Laterality=Laterality_dx--745

 select * from #final_cohort


 select f.*, e.DateValue,e.Value,e.Years,e.DisplayString,e.Laterality_dx,e.Dx_intervention_DaysDifference
 from #final_cohort f
 join #Dx_exclusion e on e.PatientDurableKey=f.PatientDurableKey
  where
      --f.Laterality= e.Laterality_dx
     f.Laterality<>'Unspecified' and  e.Laterality_dx='Unspecified'

     */

-- recover lat from dx complications
DROP TABLE IF EXISTS final_cohort_recovered; -- few recovered from dx complication ie was_inferred==2
with still_unspec as --13
(
select * from #final_cohort where Laterality='Unspecified'
), matched_pt as (
select pt.PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
```

```
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       DE.Laterality as Laterality,
       2 as was_inferred,
       ROW_NUMBER() OVER (
                   PARTITION BY pt.PatientDurableKey, DE.Laterality
                   ORDER BY DE.DateValue  DESC
               ) AS rn
from still_unspec pt
join #DxEvents_all_expanded DE
    on DE.PatientDurableKey=pt.PatientDurableKey
where DE.Laterality<>'Unspecified'  and DE.DateValue>pt.Date_retin_dx
),
--select * from matched_pt where rn=1
--order by PatientDurableKey
 removed as(
select *
from  #final_cohort c
where  not exists (
    select 1
    from matched_pt ml
    where  ml.PatientDurableKey= c.PatientDurableKey and c.Laterality='Unspecified'   )  -- and c.Laterality='Unspecified'

--c.PatientDurableKey not in (select PatientDurableKey from matched_pt) and c.Laterality='Unspecified'
),
joined as (
select PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
from removed

union all

select PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
from matched_pt where rn=1
),
ranked as (
select *,
        ROW_NUMBER() OVER (
                   PARTITION BY PatientDurableKey, Laterality
                   ORDER BY Date_retin_dx DESC
               ) AS rn
from joined
)
select PatientDurableKey,
       Date_retin_dx,
       Age_retin_dx,
       icd_retin_dx,
       DisplayString_retin_dx,
       Insurance_retin_dx,
       Laterality,
       was_inferred
into #final_cohort_recovered
from ranked where rn=1
;

/*

select COUNT_BIG(*) from #final_cohort_recovered; --6470
select  COUNT_BIG(distinct PatientDurableKey) from #final_cohort_recovered; --3904
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #final_cohort_recovered; --6470

select Laterality, count(Laterality) as N from  #final_cohort_recovered group by Laterality;
Unspecified 479
RT  3011
LT  2980
*/



DROP TABLE IF EXISTS #Dx_complication;
with MYPatients as(
select * from #final_cohort_recovered
),
expanded_DX as(
select * from #DxEvents_all_expanded
),
matched_laterality as (
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY DE.DateValue ASC) as rn_firstDx

from (select * from MYPatients ) pt --where Laterality<>'Unspecified'
inner join expanded_DX DE
    on DE.PatientDurableKey=pt.PatientDurableKey and DE.Laterality=pt.Laterality
    where DE.DateValue>pt.Date_retin_dx
    ),

match_remainder as (
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY ABS(DATEDIFF(DAY, pt.Date_retin_dx, DE.DateValue)) ASC) as rn_firstDx
```

```
from ( select * from MYPatients mp
where not exists (
    select 1
    from matched_laterality ml
    where  ml.PatientDurableKey= mp.PatientDurableKey and ml.Laterality=mp.Laterality
        )

    ) pt
inner join ( select * from expanded_DX where Laterality='Unspecified') DE --where Laterality='Unspecified'
    on DE.PatientDurableKey=pt.PatientDurableKey
    where DE.DateValue>pt.Date_retin_dx
),

Match_remainder_unspec as(
select  pt.*,
       -- DE.PatientDurableKey,
        DE.DateValue,
        DE.[Value],
        DE.Years,
        DE.DisplayString,
        DE.CoverageFinancialCLass,
        DE.Laterality as Laterality_dx,
        ROW_NUMBER() OVER(PARTITION BY pt.PatientDurableKey,pt.Laterality ORDER BY ABS(DATEDIFF(DAY, pt.Date_retin_dx, DE.DateValue)) ASC) as rn_firstDx

from ( select * from MYPatients mp
where not exists (
    select 1
    from matched_laterality ml
    where  ml.PatientDurableKey= mp.PatientDurableKey and ml.Laterality=mp.Laterality
        )
    and not exists (
    select 1
    from match_remainder m2
    where  m2.PatientDurableKey= mp.PatientDurableKey and m2.Laterality=mp.Laterality
    )
    ) pt
inner join ( select * from expanded_DX ) DE --where Laterality='Unspecified'
    on DE.PatientDurableKey=pt.PatientDurableKey
    where DE.DateValue>pt.Date_retin_dx and pt.Laterality='Unspecified'

),
joined_data as (
select * from matched_laterality where rn_firstDx=1

union all

select * from match_remainder where rn_firstDx=1


union all

select * from Match_remainder_unspec where rn_firstDx=1

)
select *,
       DATEDIFF(DAY, Date_retin_dx, DateValue) as Days_complication
into #Dx_complication
from joined_data
;


/*
select * from #Dx_complication order by PatientDurableKey,Laterality


select * from #first_sc_retin_dx where PatientDurableKey='12697099'
--select * from #Dx_exclusion where PatientDurableKey='12697099'
select * from #DxEvents_all_expanded where PatientDurableKey='12697099'


select * from #final_cohort_recovered where PatientDurableKey='12697099'
select * from #DxEvents_all_expanded where PatientDurableKey='12697099'


select COUNT_BIG(*) from #Dx_complication; --199
select  COUNT_BIG(distinct PatientDurableKey) from #Dx_complication; --174
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #Dx_complication; --199

select Laterality, count(Laterality) as N from  #Dx_complication group by Laterality;
LT  100
RT  99

select COUNT_BIG(*) from #final_cohort_recovered; --6470
select  COUNT_BIG(distinct PatientDurableKey) from #final_cohort_recovered; --3904
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #final_cohort_recovered; --6470

select Laterality, count(Laterality) as N from  #final_cohort_recovered group by Laterality;
Unspecified 479
RT  3011
LT  2980



select COUNT_BIG(*) from #final_cohort_recovered_complications; --6470
select  COUNT_BIG(distinct PatientDurableKey) from #final_cohort_recovered_complications; --3904
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #final_cohort_recovered_complications; --6470

select Laterality, count(Laterality) as N from  #final_cohort_recovered_complications group by Laterality;
Unspecified 479
RT  3011
LT  2980


select COUNT_BIG(*) from #final_cohort_recovered_complications_pt_demo; --5775
select  COUNT_BIG(distinct PatientDurableKey) from #final_cohort_recovered_complications_pt_demo; --3488
select  COUNT_BIG(distinct concat (PatientDurableKey,Laterality)) from #final_cohort_recovered_complications_pt_demo; --5775

select Laterality, count(Laterality) as N from  #final_cohort_recovered_complications_pt_demo group by Laterality;
Unspecified 432
RT  2684
LT  2659


 select COUNT_BIG(*) from #final_cohort_recovered_complications where Days_complication is not null --199
 select COUNT_BIG(*) from #final_cohort_recovered_complications_pt_demo where Days_complication is not null --173
*/
```

```sql

DROP TABLE IF EXISTS #final_cohort_recovered_complications;
select f.*,
       c.DateValue as Date_complication,
       c.Value as icd_complication,
       c.Years as age_complication,
       c.DisplayString as DisplayString_complication,
       c.CoverageFinancialClass as Insurance_complication,
       c.Laterality_dx as Laterality_complication,
       c.Days_complication
into #final_cohort_recovered_complications
from #final_cohort_recovered f
left join #Dx_complication c
    on f.PatientDurableKey=c.PatientDurableKey and c.Laterality=f.Laterality
    ;

--select * from #final_cohort_recovered_complications order by PatientDurableKey,Laterality;




DROP TABLE IF EXISTS #PatientDemo; -- 2851
select DurableKey as PatientDurableKey,
       BirthDate,
       Sex,
       MaritalStatus,
       PreferredLanguage,
        CASE
            when PreferredLanguage='English' Then 'English'
            ELSE 'Non-English'
        end as [Language],
       Ethnicity,
       FirstRace,
       --SecondRace,
       --ThirdRace,
       --FourthRace,
       --FifthRace,
       MultiRacial,
       DeathDate,
       [Status],
       Country,
       StateOrProvince,
       StateOrProvinceAbbreviation,
       --SourceCountry_X,
      -- PostalCodeKey, not avail here
        --SviOverallPctlRankingByZip2018_X [svi_overall],
        --SviHouseholdCompositionPctlRankingByZip2018_X [svi_house],
        --SviHousingTypeTransportationPctlRankingByZip2018_X [svi_transport],
        --SviMinorityStatusLanguagePctlRankingByZip2018_X [svi_minority],
        --SviSocioeconomicPctlRankingByZip2018_X [svi_ses],

        SviOverallPctlRankByZip2020_X [svi_overall_2020],
        SviSocioeconomicPctlRankByZip2020_X [svi_ses_2020],
        SviHouseholdCharacteristicsPctlRankByZip2020_X [svi_house_2020],
        --SviMinorityStatusLanguagePctlRankingByZip2020_X [svi_minority_2020],
        SviRacialEthnicMinorityStatusPctlRankByZip2020_X [svi_minority_2020],
        SviHousingTypeTransportationPctlRankByZip2020_X [svi_transport_2020] ,

       PrimaryRUCA_X [ruca]

into #PatientDemo
from PatientDim
where DurableKey in (select distinct PatientDurableKey from #final_cohort_recovered_complications)
--AND EXISTS (
--    SELECT 1
--    FROM EyeExamTestFact eye
--       WHERE PatientDim.DurableKey=eye.PatientDurableKey
--     AND eye.ExamType IN (N'VA: Visual Acuity')
--    )
 AND (SviOverallPctlRankByZip2020_X IS NOT NULL) -- OR pt.SviOverallPctlRankingByZip2018_X IS NOT NULL
       AND Sex NOT in ('Unknown','*Masked','*Unspecified','Other')
        AND Ethnicity NOT in ('Unknown','*Masked','*Unspecified')
        AND NULLIF(FirstRace, '') IS NOT NULL
        AND NULLIF(Sex, '') IS NOT NULL
        AND NULLIF(PreferredLanguage, '') IS NOT NULL
        AND NULLIF(Ethnicity, '') IS NOT NULL
;


--select count_big(distinct PatientDurableKey) from #final_cohort_recovered_complications;--3904
--select count_big(distinct PatientDurableKey) from #PatientDemo;--3488



DROP TABLE IF EXISTS #final_cohort_recovered_complications_pt_demo; -- removed missing demograaphic
with encs as (
select distinct PatientDurableKey, EncounterCountAfterFirstDate_opt,LastEncounterDate_opt,EncounterCountAfterFirstDate,LastEncounterDate from #cohort
)
select dx.*,EncounterCountAfterFirstDate_opt,LastEncounterDate_opt,EncounterCountAfterFirstDate,LastEncounterDate,
         DATEDIFF(DAY, Date_retin_dx, LastEncounterDate) AS FU_day
into #final_cohort_recovered_complications_pt_demo
from #final_cohort_recovered_complications dx
left join encs e
    on e.PatientDurableKey=dx.PatientDurableKey
where dx.PatientDurableKey in ( select PatientDurableKey from #PatientDemo) -- filter to pt with no missing demographics
;

--select * from  #final_cohort_recovered_complications_pt_demo;

/*
SELECT [column] = c.[name],
       [type] = t.[name],
       CASE WHEN t.[name] = N'nvarchar' THEN c.max_length/2 ELSE c.max_length END max_length
  FROM tempdb.sys.columns AS c
    INNER JOIN tempdb.sys.types AS t
      ON c.system_type_id = t.system_type_id
      AND t.system_type_id = t.user_type_id
  WHERE [object_id] = OBJECT_ID( N'tempdb.dbo.#final_cohort_recovered_complications_pt_demo');


  12/5/2025

    INSERT INTO PROJECTS.ProjectD821712.dbo.PatientDemo
    select * from #PatientDemo;
```

```
   INSERT INTO PROJECTS.ProjectD821712.dbo.final_cohort
  select * from #final_cohort_recovered_complications_pt_demo;


*/
```

```
Use COSMOS;

---------------- update missing unspec.

/*
DROP TABLE IF EXISTS #cohort;
select *
into #cohort
from PROJECTS.ProjectD821712.dbo.final_cohort;


DROP TABLE IF EXISTS #inferred_lat;
with unspec_cohort as (
select * from #cohort where Laterality='Unspecified'
),
match_pt as (
select c.PatientDurableKey, c.Laterality,r.Laterality as Laterality2,
 ROW_NUMBER() OVER (
                  PARTITION BY c.PatientDurableKey, c.Laterality
                  ORDER BY r.Date_retin_dx ASC
              ) AS rn

from unspec_cohort c
join  (select * from  PROJECTS.ProjectD821712.dbo.SC_retin_dx where Laterality<>'Unspecified') r
on c.PatientDurableKey=r.PatientDurableKey
)

select *
into #inferred_lat
from match_pt where rn=1
;

UPDATE d
SET
    d.Laterality = i.Laterality2,
    d.was_inferred = 1
FROM PROJECTS.ProjectD821712.dbo.final_cohort d
JOIN #inferred_lat i
    ON d.PatientDurableKey = i.PatientDurableKey
WHERE d.Laterality = 'Unspecified';




UPDATE d
SET
    d.Laterality = i.Laterality2,
    d.was_inferred = 1
FROM PROJECTS.ProjectD821712.dbo.final_cohort_meds d
JOIN #inferred_lat i
    ON d.PatientDurableKey = i.PatientDurableKey
WHERE d.Laterality = 'Unspecified';


--select Laterality, count(*) as N from PROJECTS.ProjectD821712.dbo.final_cohort group by Laterality;
--Unspecified   405
--RT    2699
--LT    2671

--select Laterality, count(*) as N from PROJECTS.ProjectD821712.dbo.final_cohort_meds  group by Laterality;
--Unspecified   405
--RT    2699
--LT    2671

*/


Use COSMOS;

DROP TABLE IF EXISTS #cohort;
```

```sql
with ranked as (
select *, ROW_NUMBER() OVER(PARTITION BY PatientDurableKey,Laterality ORDER BY Date_retin_dx ASC) as rn
from PROJECTS.ProjectD821712.dbo.final_cohort
)
select *
into #cohort
from ranked
where rn=1
;


--  ##############################


DROP TABLE IF EXISTS #Meds;
with all_hydrox as( -- get keys
select DISTINCT MedicationKey,
    PharmaceuticalClass,
    SimpleGenericName,
    TherapeuticClass
  FROM MedicationDim
  where SimpleGenericName like '%hydroxyurea%'
        And PharmaceuticalClass like '%SICKLE CELL%'
),
OuterOrders AS (
    SELECT
        orders.MedicationKey,
        orders.PatientDurableKey,
         orders.RefillsWritten,orders.DiscontinueReason,orders.FirstIndicationForUse,orders.SecondIndicationForUse,orders.ThirdIndicationForUse,orders.FourthIndicationForUse,orders.Frequency,
        startDateInfo.DateValue AS StartDate,
        endDateInfo.DateValue AS EndDate,
        1 AS used_hydroxyurea,
        MRCF.StartDateKey, MRCF.EndDateKey, MRCF.ComponentType_X,MRCF.Frequency_X
    FROM (select PatientDurableKey, min(Date_retin_dx) as Date_retin_dx from #cohort group by PatientDurableKey) pt
    JOIN MedicationOrderFact orders
        ON orders.PatientDurableKey = pt.PatientDurableKey
    INNER JOIN all_hydrox meds
        ON orders.MedicationKey = meds.MedicationKey
    INNER JOIN DateDim startDateInfo
        ON startDateInfo.DateKey = orders.StartDateKey
    INNER JOIN DateDim endDateInfo
        ON endDateInfo.DateKey = orders.EndDateKey
    INNER JOIN MedicationOrderComponentFact MRCF  on MRCF.MedicationOrderKey= orders.MedicationOrderKey
    --WHERE startDateInfo.DateValue < pt.proc_date
)
select *
into #Meds
from OuterOrders

;

/*

select * from #Meds order by PatientDurableKey, StartDate;


select count_big(distinct PatientDurableKey) from #Meds

select count_big(distinct PatientDurableKey) from #Meds where StartDate is null;
select count_big(distinct PatientDurableKey) from #Meds where EndDate is null;

*/

DROP TABLE IF EXISTS #med_clean;
with step1 as (

SELECT
    PatientDurableKey,

    CASE
        WHEN StartDate IS NULL AND EndDate IS NOT NULL
        THEN DATEADD(day,-90,EndDate)
        ELSE StartDate
    END AS StartDate,

    CASE
        WHEN EndDate IS NULL AND StartDate IS NOT NULL
        THEN DATEADD(day,90,StartDate)
        WHEN YEAR(EndDate)=9999 THEN DATEADD(day,90,StartDate)
        ELSE EndDate
    END AS EndDate


FROM #Meds
WHERE NOT (StartDate IS NULL AND EndDate IS NULL)
),
step2 as (
select
PatientDurableKey,StartDate,
CASE
    when StartDate > EndDate
        THEN DATEADD(day,90,StartDate)
        ELSE EndDate
        END AS EndDate
from step1
)
SELECT
PatientDurableKey,
StartDate,
MAX(EndDate) AS EndDate
INTO #med_clean
FROM step2
GROUP BY PatientDurableKey, StartDate
;

--select top 100 * from #med_clean


DROP TABLE IF EXISTS #baseline_exposure;
SELECT
c.PatientDurableKey,
c.Laterality,
c.Date_retin_dx,

CASE
WHEN EXISTS (
```

```
    SELECT 1
    FROM #med_clean m
    WHERE m.PatientDurableKey = c.PatientDurableKey
    AND (
        m.StartDate <= c.Date_retin_dx
        OR m.EndDate <= c.Date_retin_dx
    )
)
THEN 1
ELSE 0
END AS exposed_at_index

INTO #baseline_exposure
FROM #cohort c;



DROP TABLE IF EXISTS #post_index_start;
SELECT
c.PatientDurableKey,
MIN(m.StartDate) AS first_start_after_index

INTO #post_index_start

FROM #cohort c
JOIN #med_clean m
ON c.PatientDurableKey = m.PatientDurableKey

WHERE m.StartDate > c.Date_retin_dx

GROUP BY c.PatientDurableKey;



DROP TABLE IF EXISTS #exposure_timeline;

SELECT
b.PatientDurableKey,
b.Laterality,
b.Date_retin_dx,

b.exposed_at_index,

p.first_start_after_index,

CASE
WHEN b.exposed_at_index = 1 THEN b.Date_retin_dx
WHEN p.first_start_after_index IS NOT NULL THEN p.first_start_after_index
ELSE NULL
END AS exposure_start_date

INTO #exposure_timeline

FROM #baseline_exposure b
LEFT JOIN #post_index_start p
ON b.PatientDurableKey = p.PatientDurableKey;


--select * from #exposure_timeline






---build complication table first


DROP TABLE IF EXISTS #dx_complications_followup;
SELECT
c.PatientDurableKey,
c.Laterality,
d.icd10 as code,
d.DateValue as Date_complication
INTO #dx_complications_followup
FROM #cohort c
JOIN PROJECTS.ProjectD821712.dbo.Dx_complication_long d
ON c.PatientDurableKey = d.PatientDurableKey
AND (
    c.Laterality = d.Laterality
    OR d.Laterality = 'Unspecified'
    OR c.Laterality = 'Unspecified'
)
WHERE d.DateValue > c.Date_retin_dx
;

/*

select PatientDurableKey from #dx_complications_followup group by PatientDurableKey having count(*)>6;

select * from #dx_complications_followup where PatientDurableKey='447650088'


select PatientDurableKey from PROJECTS.ProjectD821712.dbo.Dx_complication_long group by PatientDurableKey having count(*)>25;

select * from  PROJECTS.ProjectD821712.dbo.Dx_complication_long where PatientDurableKey='59996246';
select * from  #cohort where PatientDurableKey='59996246';

select * from  PROJECTS.ProjectD821712.dbo.SC_retin_dx where PatientDurableKey='59996246';


--select * from PROJECTS.ProjectD821712.dbo.injection_complication_long order by PatientDurableKey,Laterality;

select * from PROJECTS.ProjectD821712.dbo.Dx_complication_long where
PatientDurableKey in (select PatientDurableKey from #cohort)

--select Laterality, count(*) as N from #cohort group by Laterality;
--Unspecified   432
--RT    2684
--LT    2659
--select Laterality, count(*) as N from PROJECTS.ProjectD821712.dbo.Dx_complication_long group by Laterality;
--Unspecified   8
--RT    1243
--LT    1300
```

```sql
select PatientDurableKey from #cohort group by PatientDurableKey having count(*)>2;

select count(distinct concat(PatientDurableKey,Laterality)) from #cohort;
select count(distinct concat(PatientDurableKey,Laterality)) from PROJECTS.ProjectD821712.dbo.Dx_complication_long;

select * from #cohort where PatientDurableKey='9275001';


SELECT COUNT(*)
FROM #cohort c
JOIN  PROJECTS.ProjectD821712.dbo.Dx_complication_long d
  ON c.PatientDurableKey = d.PatientDurableKey;

  SELECT COUNT(*)
FROM #cohort c
JOIN PROJECTS.ProjectD821712.dbo.Dx_complication_long d
  ON c.PatientDurableKey = d.PatientDurableKey
 AND (
      c.Laterality = d.Laterality
      OR d.Laterality = 'Unspecified'
      OR c.Laterality = 'Unspecified'
 );

 SELECT
    c.Laterality AS cohort_lat,
    d.Laterality AS comp_lat,
    COUNT(*) AS cnt
FROM #cohort c
JOIN PROJECTS.ProjectD821712.dbo.Dx_complication_long d
  ON c.PatientDurableKey = d.PatientDurableKey
GROUP BY c.Laterality, d.Laterality
ORDER BY cnt DESC;

*/


DROP TABLE IF EXISTS #sx_complications_followup;

SELECT
c.PatientDurableKey,
c.Laterality,
s.CptCode as code,
s.complication_sx_date as Date_complication
INTO #sx_complications_followup
FROM #cohort c
JOIN PROJECTS.ProjectD821712.dbo.sx_complication_long s
ON c.PatientDurableKey = s.PatientDurableKey
AND (
    c.Laterality = s.Laterality
    OR s.Laterality = 'Unspecified'
    OR c.Laterality = 'Unspecified'
)
WHERE s.complication_sx_date > c.Date_retin_dx;


DROP TABLE IF EXISTS #inj_complications_followup;

SELECT
c.PatientDurableKey,
c.Laterality,
i.CptCode as code,
i.injection_date as Date_complication
INTO #inj_complications_followup
FROM #cohort c
JOIN PROJECTS.ProjectD821712.dbo.injection_complication_long  i
ON c.PatientDurableKey = i.PatientDurableKey
AND  (
    c.Laterality = i.Laterality
    OR i.Laterality = 'Unspecified'
    OR c.Laterality = 'Unspecified'
)
WHERE i.injection_date > c.Date_retin_dx;


------------ big table
DROP TABLE IF EXISTS #all_complications;
with all_complications as (
SELECT
PatientDurableKey,
Laterality,
Date_complication,
code,
'dx' AS source
FROM #dx_complications_followup

UNION ALL

SELECT
PatientDurableKey,
Laterality,
Date_complication,
code,
'sx'
FROM #sx_complications_followup

UNION ALL

SELECT
PatientDurableKey,
Laterality,
Date_complication,
code,
'injection'
FROM #inj_complications_followup
)
select *
into #all_complications
from all_complications
;



--DROP TABLE IF EXISTS #hydroxyurea_episodes_lag;
--SELECT
--PatientDurableKey,
--episode_id,
```

```sql
--episode_start,
--episode_end,
--DATEADD(day,90,episode_start) AS exposure_start,
--DATEADD(day,90,episode_end)   AS exposure_end

--INTO #hydroxyurea_episodes_lag
--FROM #hydroxyurea_episodes;

select * from PROJECTS.ProjectD821712.dbo.final_events


DROP TABLE IF EXISTS #complications_exposure;

SELECT
c.*,

CASE
WHEN EXISTS (

SELECT 1
FROM #exposure_timeline e

WHERE e.PatientDurableKey = c.PatientDurableKey
AND e.exposure_start_date is not null
AND c.Date_complication >= DATEADD(day,90,e.exposure_start_date) --e.exposure_start_date
)
THEN 1
ELSE 0
END AS hydroxyurea_exposed_event
INTO #complications_exposure
FROM #all_complications c;

--select * from #complications_exposure;


ALTER TABLE #complications_exposure
ADD complication_type VARCHAR(30);
GO

UPDATE #complications_exposure
SET complication_type =
CASE
WHEN code LIKE 'H43.1%' THEN 'vitreous_hemorrhage'
WHEN code LIKE 'H33.4%' THEN 'tractional_rd'
WHEN code = '67228' THEN 'PRP'
WHEN code IN ('67036','67039','67040','67041','67042','67043','67107','67108','67112','67113') THEN 'vitrectomy'
WHEN code = '67028' THEN 'injection'
END;

--select * from #complications_exposure

--select count(*) from #complications_exposure where hydroxyurea_exposed_event=1


DROP TABLE IF EXISTS #final_events;

SELECT
c.PatientDurableKey,
c.Laterality,
c.Date_retin_dx,

e.Date_complication,
e.code,
e.complication_type,
e.hydroxyurea_exposed_event,

et.exposed_at_index,
et.first_start_after_index,
et.exposure_start_date

INTO #final_events

FROM #cohort c
LEFT JOIN #complications_exposure e
ON c.PatientDurableKey = e.PatientDurableKey
AND c.Laterality = e.Laterality
AND e.Date_complication > c.Date_retin_dx
left join #exposure_timeline et
on et.PatientDurableKey=c.PatientDurableKey
AND et.Laterality=c.Laterality

;


--select count(*) from #final_events where hydroxyurea_exposed_event=1

--select * from  #complications_exposure


--select * from  #final_events where Date_complication is null;

-- select count(*) from #final_events --6489


/*
SELECT [column] = c.[name],
       [type] = t.[name],
       CASE WHEN t.[name] = N'nvarchar' THEN c.max_length/2 ELSE c.max_length END max_length
  FROM tempdb.sys.columns AS c
    INNER JOIN tempdb.sys.types AS t
      ON c.system_type_id = t.system_type_id
      AND t.system_type_id = t.user_type_id
  WHERE [object_id] = OBJECT_ID( N'tempdb.dbo.#final_events');

  --- SAVE 05/26/2026,  04/20/2026

   INSERT INTO PROJECTS.ProjectD821712.dbo.final_events
   select * from #final_events;

*/



---------------------------------------------------------------- Progression
```

```sql
DROP TABLE IF EXISTS #long_dx;
select *
into #long_dx
from  PROJECTS.ProjectD821712.dbo.SC_retin_dx
;


DROP TABLE IF EXISTS #lat_flags;

SELECT
PatientDurableKey,
MAX(CASE WHEN Laterality = 'RT' THEN 1 ELSE 0 END) AS has_RT,
MAX(CASE WHEN Laterality = 'LT' THEN 1 ELSE 0 END) AS has_LT
INTO #lat_flags
FROM #long_dx
GROUP BY PatientDurableKey;




DROP TABLE IF EXISTS #dx_expanded;

SELECT
d.PatientDurableKey,
d.Date_retin_dx,
d.icd_retin_dx,
d.DisplayString_retin_dx,
d.Laterality
INTO #dx_expanded
FROM #long_dx d
JOIN #lat_flags f
ON d.PatientDurableKey = f.PatientDurableKey

WHERE d.Laterality IN ('RT','LT')

UNION ALL

-- only expand unspecified if NO known laterality exists
SELECT
d.PatientDurableKey,
d.Date_retin_dx,
d.icd_retin_dx,
d.DisplayString_retin_dx,
'RT'
FROM #long_dx d
JOIN #lat_flags f
ON d.PatientDurableKey = f.PatientDurableKey
WHERE d.Laterality = 'Unspecified'
AND f.has_RT = 0 AND f.has_LT = 0

UNION ALL

SELECT
d.PatientDurableKey,
d.Date_retin_dx,
d.icd_retin_dx,
d.DisplayString_retin_dx,
'LT'
FROM #long_dx d
JOIN #lat_flags f
ON d.PatientDurableKey = f.PatientDurableKey
WHERE d.Laterality = 'Unspecified'
AND f.has_RT = 0 AND f.has_LT = 0;




DROP TABLE IF EXISTS #first_eye_dx;

WITH ranked AS (

SELECT
PatientDurableKey,
Laterality,
Date_retin_dx,
icd_retin_dx,

ROW_NUMBER() OVER (
PARTITION BY PatientDurableKey, Laterality
ORDER BY Date_retin_dx ASC
) AS rn

FROM #dx_expanded
)

SELECT
PatientDurableKey,
Laterality,
Date_retin_dx AS first_dx_date,
icd_retin_dx AS first_icd

INTO #first_eye_dx

FROM ranked
WHERE rn = 1;

DROP TABLE IF EXISTS #eligible_progression_eyes;

SELECT *
INTO #eligible_progression_eyes
FROM #first_eye_dx
WHERE first_icd LIKE 'H36.81%';



DROP TABLE IF EXISTS #dx_ordered;

SELECT
PatientDurableKey,
Laterality,
Date_retin_dx,

CASE
WHEN icd_retin_dx LIKE 'H36.82%' THEN 1  -- PDR
WHEN icd_retin_dx LIKE 'H36.81%' THEN 0  -- NPDR
END AS is_PDR,
```

```
ROW_NUMBER() OVER (
PARTITION BY PatientDurableKey, Laterality
ORDER BY Date_retin_dx
) AS visit_order

INTO #dx_ordered
FROM #dx_expanded;


DROP TABLE IF EXISTS #dx_progression_cohort;

SELECT d.*
INTO #dx_progression_cohort
FROM #dx_ordered d
JOIN #eligible_progression_eyes e
ON d.PatientDurableKey = e.PatientDurableKey
AND d.Laterality = e.Laterality;

--select * from #dx_progression_cohort order by PatientDurableKey,Laterality;


DROP TABLE IF EXISTS #progression_flags;

WITH seq AS (
SELECT
*,

LAG(is_PDR,1) OVER (
PARTITION BY PatientDurableKey, Laterality
ORDER BY Date_retin_dx
) AS prev1,

LEAD(is_PDR,1) OVER (
PARTITION BY PatientDurableKey, Laterality
ORDER BY Date_retin_dx
) AS next1

FROM #dx_progression_cohort
)

SELECT
PatientDurableKey,
Laterality,
Date_retin_dx,

CASE
WHEN is_PDR = 1
AND next1 = 1      -- second confirming visit
AND prev1 = 0      -- came from NPDR
THEN 1
ELSE 0
END AS confirmed_progression

INTO #progression_flags
FROM seq;


DROP TABLE IF EXISTS #progression_eye;

SELECT
p.PatientDurableKey,
p.Laterality,

MIN(p.Date_retin_dx) AS progression_date

INTO #progression_eye
FROM #progression_flags p
WHERE p.confirmed_progression = 1
GROUP BY
p.PatientDurableKey,
p.Laterality;


DROP TABLE IF EXISTS #first_npdr;

SELECT
PatientDurableKey,
Laterality,
MIN(Date_retin_dx) AS first_npdr_date
INTO #first_npdr
FROM #dx_ordered
WHERE is_PDR = 0
GROUP BY PatientDurableKey, Laterality;



DROP TABLE IF EXISTS #progression_final;

SELECT
n.PatientDurableKey,
n.Laterality,
n.first_npdr_date,
p.progression_date,

DATEDIFF(day, n.first_npdr_date, p.progression_date) AS days_to_progression,

CASE
WHEN p.progression_date IS NOT NULL THEN 1
ELSE 0
END AS progressed_flag

INTO #progression_final

FROM #first_npdr n
LEFT JOIN #progression_eye p
ON n.PatientDurableKey = p.PatientDurableKey
AND n.Laterality = p.Laterality;


--select count(*) from #progression_final where progressed_flag=1;




DROP TABLE IF EXISTS #progression_patient;
```

```sql
SELECT
PatientDurableKey,

MIN(progression_date) AS patient_progression_date,

MIN(days_to_progression) AS patient_days_to_progression,

MAX(progressed_flag) AS any_progression

INTO #progression_patient

FROM #progression_final
GROUP BY PatientDurableKey;

/*
select * from  #lat_flags;

select Laterality, count(*) as N from #long_dx group by Laterality;
Unspecified 1946
RT  12906
LT  12839

select count(*) from #long_dx --27691
select count(*) from #dx_expanded --28709

select * from #dx_ordered order by Date_retin_dx;


select count(*) from #progression_final WHERE progressed_flag = 1;


select count(*) from #progression_patient WHERE any_progression = 1;
select * from #progression_patient ;


------------------------------checks --------------------

with progressed_pt as (
SELECT *
FROM #progression_final
WHERE progressed_flag = 1

)
select * from #long_dx
where PatientDurableKey in (select PatientDurableKey from progressed_pt)
ORDER BY PatientDurableKey, Laterality,Date_retin_dx
;


select * from #progression_final where PatientDurableKey='102847986';
select * from #long_dx where PatientDurableKey='102847986' order by Laterality,Date_retin_dx;

select * from #cohort where PatientDurableKey='102847986' order by Laterality,Date_retin_dx;

select * from PROJECTS.ProjectD821712.dbo.final_cohort_meds where PatientDurableKey='98899968'

*/


/*
SELECT [column] = c.[name],
       [type] = t.[name],
       CASE WHEN t.[name] = N'nvarchar' THEN c.max_length/2 ELSE c.max_length END max_length
  FROM tempdb.sys.columns AS c
    INNER JOIN tempdb.sys.types AS t
      ON c.system_type_id = t.system_type_id
      AND t.system_type_id = t.user_type_id
  WHERE [object_id] = OBJECT_ID( N'tempdb.dbo.#progression_final');

  --- SAVE 04/30/2026

   INSERT INTO PROJECTS.ProjectD821712.dbo.progression
   select * from #progression_final;

*/


-- update old table with correct dates and first dx from final_cohort


Use COSMOS;

DROP TABLE IF EXISTS #cohort;
with ranked as (
select *, ROW_NUMBER() OVER(PARTITION BY PatientDurableKey,Laterality ORDER BY Date_retin_dx ASC) as rn
from PROJECTS.ProjectD821712.dbo.final_cohort
)
select *
into #cohort
from ranked
where rn=1
;



UPDATE d
SET
    d.Date_retin_dx = i.Date_retin_dx,
    d.Age_retin_dx = i.Age_retin_dx,
    d.icd_retin_dx = i.icd_retin_dx,
    d.DisplayString_retin_dx = i.DisplayString_retin_dx

FROM PROJECTS.ProjectD821712.dbo.final_cohort_meds d
JOIN #cohort i
    ON d.PatientDurableKey = i.PatientDurableKey
    and d.Laterality = i.Laterality
    ;
```

```sql
use ProjectD821712;

drop table if exists dbo.progression
create table dbo.progression
(
PatientDurableKey   bigint,
Laterality  varchar(11),
first_npdr_date date,
progression_date    date,
days_to_progression int,
progressed_flag int
)


drop table if exists dbo.final_events
create table dbo.final_events
(
PatientDurableKey   bigint,
Laterality  varchar(11),
Date_retin_dx   date,
Date_complication   date,
icd_code    nvarchar(300),
complication_type   varchar(30),
hydroxyurea_exposed_event   int,
exposed_at_index    int,
first_start_after_index date,
exposure_start_date date
)


--select complication_type, count(*) as N from dbo.final_events group by complication_type;


----------------------------------------

drop table if exists dbo.injection_complication_long
create table dbo.injection_complication_long
(
PatientDurableKey   bigint,
CptCode nvarchar(50),
injection_date  date,
Laterality  varchar(2)
)

drop table if exists dbo.sx_complication_long
create table dbo.sx_complication_long
(
PatientDurableKey   bigint,
CptCode nvarchar(50),
complication_sx_date    date,
Laterality  varchar(2)
)


drop table if exists dbo.va_timepoints
create table dbo.va_timepoints
```

```
(
PatientDurableKey   bigint,
Laterality  varchar(11),
IndexDate   date,
baseline_va numeric(18,2),
va_3m   numeric(18,2),
va_6m   numeric(18,2),
va_9m   numeric(18,2),
va_1y   numeric(18,2),
va_2y   numeric(18,2),
va_3y   numeric(18,2),
va_5y   numeric(18,2)
)



use ProjectD821712;

drop table if exists dbo.Dx_complication_long
create table dbo.Dx_complication_long
(
PatientDurableKey   bigint,
EncounterKey    bigint,
DateValue   date,
icd10   nvarchar(300),
Years   int,
DisplayString   nvarchar(300),
CoverageFinancialClass  nvarchar(300),
Laterality  varchar(11),
PrevDate    date,
NewEpisodeFlag  int,
EpisodeNumber   int
)




drop table if exists dbo.SC_retin_dx
create table dbo.SC_retin_dx
(
PatientDurableKey   bigint,
Date_retin_dx   date,
Age_retin_dx    int,
icd_retin_dx    nvarchar(300),
DisplayString_retin_dx  nvarchar(300),
Insurance_retin_dx  nvarchar(300),
Laterality  varchar(11),
was_inferred    int,
EncounterCountAfterFirstDate_opt    int,
LastEncounterDate_opt   date,
EncounterCountAfterFirstDate    int,
LastEncounterDate   date,
FU_day_opt  int,
FU_day  int
)

--select top 1000* from dbo.SC_retin_dx order by PatientDurableKey,Laterality;

--select count(*) from dbo.SC_retin_dx --27691

drop table if exists dbo.SC_pt
create table dbo.SC_pt
(
PatientDurableKey   bigint,
EncounterKey    bigint,
Date_SC_dx  date,
Age_SC_dx   int,
icd_SC_dx   nvarchar(300),
DisplayString   nvarchar(300),
Genotype    varchar(5),
Insurance_sc_dx nvarchar(300)
)




drop table if exists dbo.final_cohort_meds
create table dbo.final_cohort_meds
```

```
(
PatientDurableKey   bigint,
Date_retin_dx   date,
Age_retin_dx    int,
icd_retin_dx    nvarchar(300),
DisplayString_retin_dx  nvarchar(300),
Insurance_retin_dx  nvarchar(300),
Laterality  varchar(11),
was_inferred    int,
Date_complication   date,
icd_complication    nvarchar(300),
age_complication    int,
DisplayString_complication  nvarchar(300),
Insurance_complication  nvarchar(300),
Laterality_complication varchar(11),
Days_complication   int,
EncounterCountAfterFirstDate_opt    int,
LastEncounterDate_opt   date,
EncounterCountAfterFirstDate    int,
LastEncounterDate   date,
FU_day  int,
startdate_hydroxyurea   date,
endDate_hydroxyurea date,
Date_SC_dx  date,
Age_SC_dx   int,
icd_SC_dx   nvarchar(300),
DisplayString   nvarchar(300),
Genotype    varchar(5),
Insurance_sc_dx nvarchar(300),
exposed_group   varchar(18),
days_sc_dx_to_hydrox    int,
days_retin_dx_to_hydrox int
)




use ProjectD821712;

drop table if exists dbo.PatientDemo
create table dbo.PatientDemo
(
PatientDurableKey   bigint,
BirthDate   date,
Sex nvarchar(300),
MaritalStatus   nvarchar(300),
PreferredLanguage   nvarchar(300),
[Language] nvarchar(11),
Ethnicity   nvarchar(300),
FirstRace   nvarchar(300),
MultiRacial tinyint,
DeathDate   date,
[Status]    nvarchar(300),
Country nvarchar(300),
StateOrProvince nvarchar(300),
StateOrProvinceAbbreviation nvarchar(300),
--SourceCountry_X   nvarchar(300),
--svi_overall   numeric(10,4),
--svi_house numeric(10,4),
--svi_transport numeric(10,4),
--svi_minority  numeric(10,4),
--svi_ses   numeric(10,4),
svi_overall_2020    numeric(6,4),
svi_ses_2020    numeric(6,4),
svi_house_2020  numeric(6,4),
svi_minority_2020   numeric(6,4),
svi_transport_2020  numeric(6,4),
ruca    nvarchar(50)
)




drop table if exists dbo.final_cohort
create table dbo.final_cohort
(
PatientDurableKey   bigint,
Date_retin_dx   date,
Age_retin_dx    int,
```

```
icd_retin_dx    nvarchar(300),
DisplayString_retin_dx  nvarchar(300),
Insurance_retin_dx  nvarchar(300),
Laterality  varchar(11),
was_inferred    int,
Date_complication   date,
icd_complication    nvarchar(300),
age_complication    int,
DisplayString_complication  nvarchar(300),
Insurance_complication  nvarchar(300),
Laterality_complication varchar(11),
Days_complication   int,
EncounterCountAfterFirstDate_opt    int,
LastEncounterDate_opt   date,
EncounterCountAfterFirstDate    int,
LastEncounterDate   date,
FU_day  int
)




------------------------------




--drop table if exists dbo.Failures
--create table dbo.Failures
--(
--PatientDurableKey bigint,
--Laterality    nvarchar(50),
--high_iop_failure_days int,
--low_iop_failure_days  int,
--reduction_failure_days    int,
--nlp_failure_days  int,
--reoperation_grp   varchar(21),
--reop_failure_code nvarchar(50),
--reop_failure_days int
--)



--drop table if exists dbo.Final_cohort -- Eyes that had confirmed dx and sx, baseline iop, medical history,concurrent cataract, FU
--create table dbo.Final_cohort
--(
--PatientDurableKey bigint,
--Laterality    nvarchar(50),
--DiagnosisEventKey bigint,
--DxDate    date,
--DxAge int,
--dx_icd    nvarchar(300),
--DisplayString nvarchar(300),
--Dx_sx_days    int,
--EcounterKey_proc  bigint,
--proc_date date,
--CptCode   nvarchar(50),
--AgeAtProcedure    int,
--PostOp_date   date,
--PostOp_cpt    nvarchar(50),
--AgeAtPostOp   int,
--PostOP_treatment  varchar(30),
--Sx_PostopSX_days  int,
--baseline_iop  numeric(18,2),
--baseline_iop_date date,
--interval_sx_iop   int,
--PostOp_iop_date   date,
--PostOp_iop    numeric(18,2),
--interval_PostOp_iop   int,
--concurrent_cataract_gonio_date    date,
--concurrent_cataract_gonio_code    nvarchar(50),
--concurrent_cataract_PostOp_date   date,
--concurrent_cataract_PostOp_code   nvarchar(50),
--glaucoma_type varchar(50),
```

```
--severity  varchar(20),
--anticoagulant_date    date,
--antiplatelet_date date,
--Had_Anticoagulant int,
--Had_Antiplatelet  int,
--LastEncounterDate date,
--EncounterCountAfter   int,
--FU_Month  int
--)




--drop table if exists dbo.cohort_dx_sx -- confirmed dx and sx eyes
--create table dbo.cohort_dx_sx
--(
--PatientDurableKey bigint,
--Laterality    nvarchar(50),
--DiagnosisEventKey bigint,
--DxDate    date,
--DxAge int,
--dx_icd    nvarchar(300),
--DisplayString nvarchar(300),
--Dx_sx_days    int,
--EcounterKey_proc  bigint,
--proc_date date,
--CptCode   nvarchar(50),
--AgeAtProcedure    int

--)
```